# One blend → two sources and a smooth mixing map

Training-free deblending with pretrained diffusion priors

## A THE PIPELINE

**Input**

Blend y
Prompts $c_1$, $c_2$

**1 Separate semantics**

Text-guided latent diffusion
3 seeds → 3 candidate pairs

**2 Select**

PickScore + prompts
No ground truth

**3 Refine detail**

Pixel-space diffusion
Refine selected pair

**Recover**

Source 1 + source 2
Alpha map

## B INSIDE JOINT SAMPLING

Shown for Stage 1; both stages use joint data consistency

**Two branches · one shared pretrained prior**

**Source 1**

+ $c_1$ / − $c_2$

**Source 2**

+ $c_2$ / − $c_1$

Cross-negative prompts discourage source leakage

**Decode estimates and reblend**

$$\hat{y} = \alpha_\phi \odot \hat{x}_1 + (1 - \alpha_\phi) \odot \hat{x}_2$$

Compare reconstructed blend with y

**Correct together**

Update both source samples

Fit smooth α using 6 coefficients

Smooth α discourages edge leakage into the mask